# Knowledge Graphs

## Lecture 5 - Knowledge Graph Applications

## 5.6 Knowledge Graph Analytics

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# 5.6 Knowledge Graph Analytics

# Knowledge Mining and Knowledge Discovery

**Knowledge Discovery**

**valid novel potentially useful**

**ultimately understandable patterns**

- **valid**: to a certain degree the discovered patterns should also hold for new, previously unseen problem instances.
- **novel**: at least to the system and preferable to the user.
- **potentially useful**: they should lead to some benefit to the user or task.
- **ultimately understandable**: the end user should be able to interpret the patterns either immediately or after some post-processing.

# Knowledge Mining and Knowledge Discovery

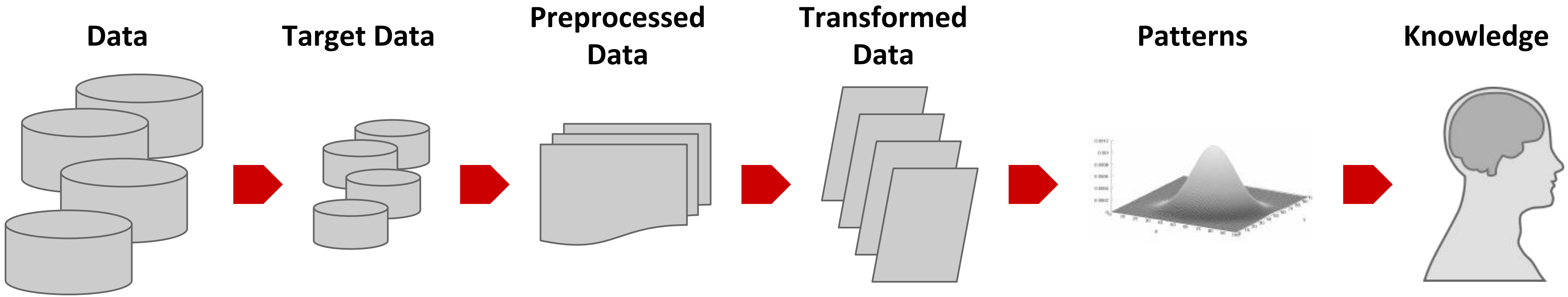
**Knowledge Discovery**

**valid novel potentially useful**

**ultimately understandable patterns**

- **Goals:**
  - **Descriptive Modelling:** explains the characteristic and the behaviour of the observed data
  - **Predictive Modelling:** predicts the behaviour of new data based on some model
- **Important:**
  - The extracted model/pattern does not have to apply in 100% of the cases

# The Knowledge Discovery Process




**Selection:**

**Preprocessing/ Cleaning:**

**Transformation:**

**Data Mining:**

**Evaluation:**

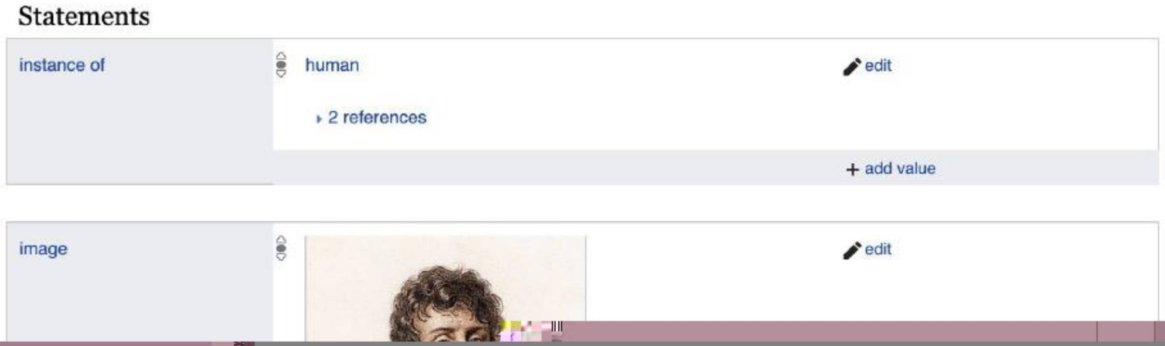
# Hands-On Example - Knowledge Graph Analytics

- **Physicists**
- **DBpedia** **Wikidata**

- **Data Acquisition**
  - **random example**

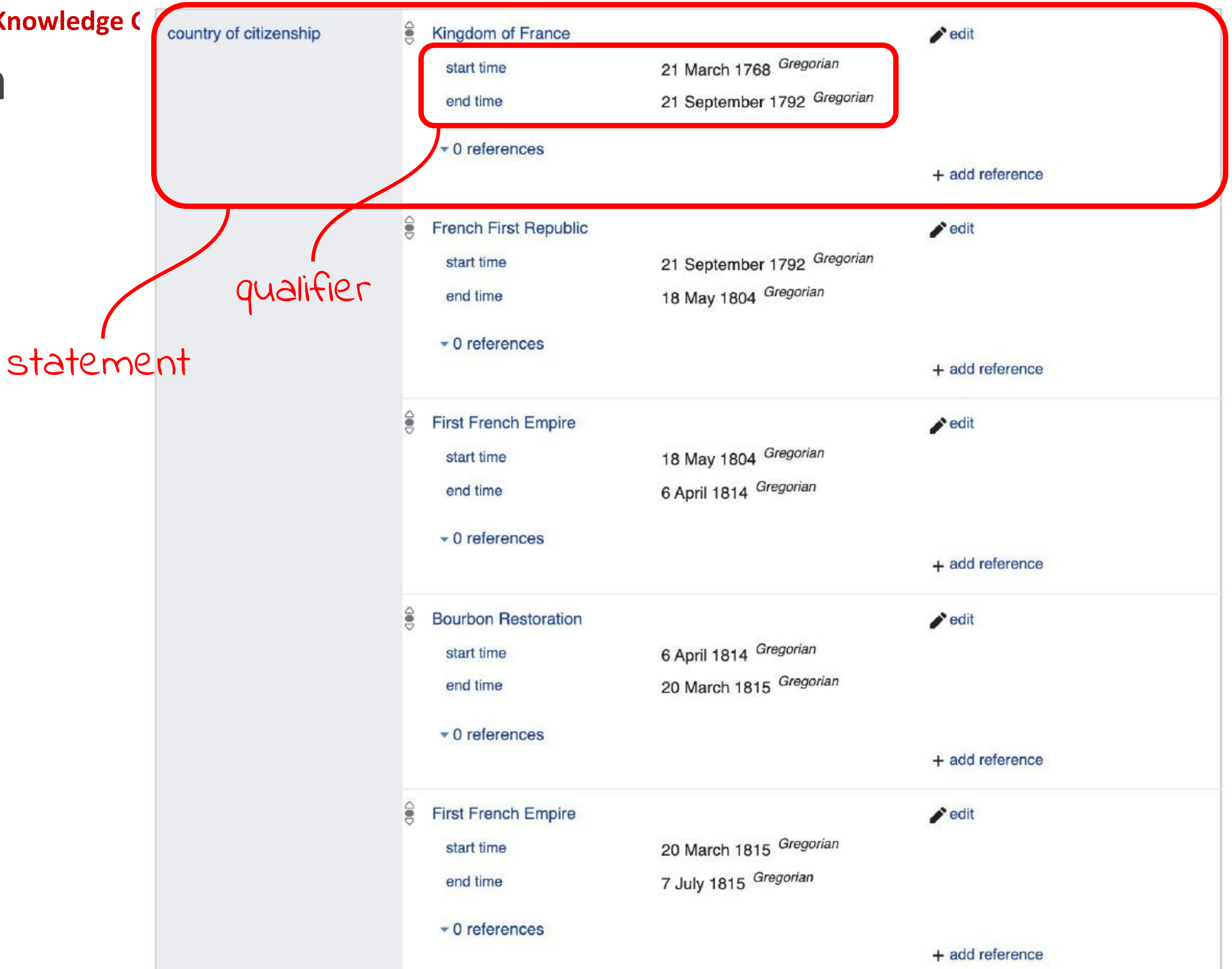
# 1. Data Acquisition

- random example



# 1. Data Acquisition

- random example

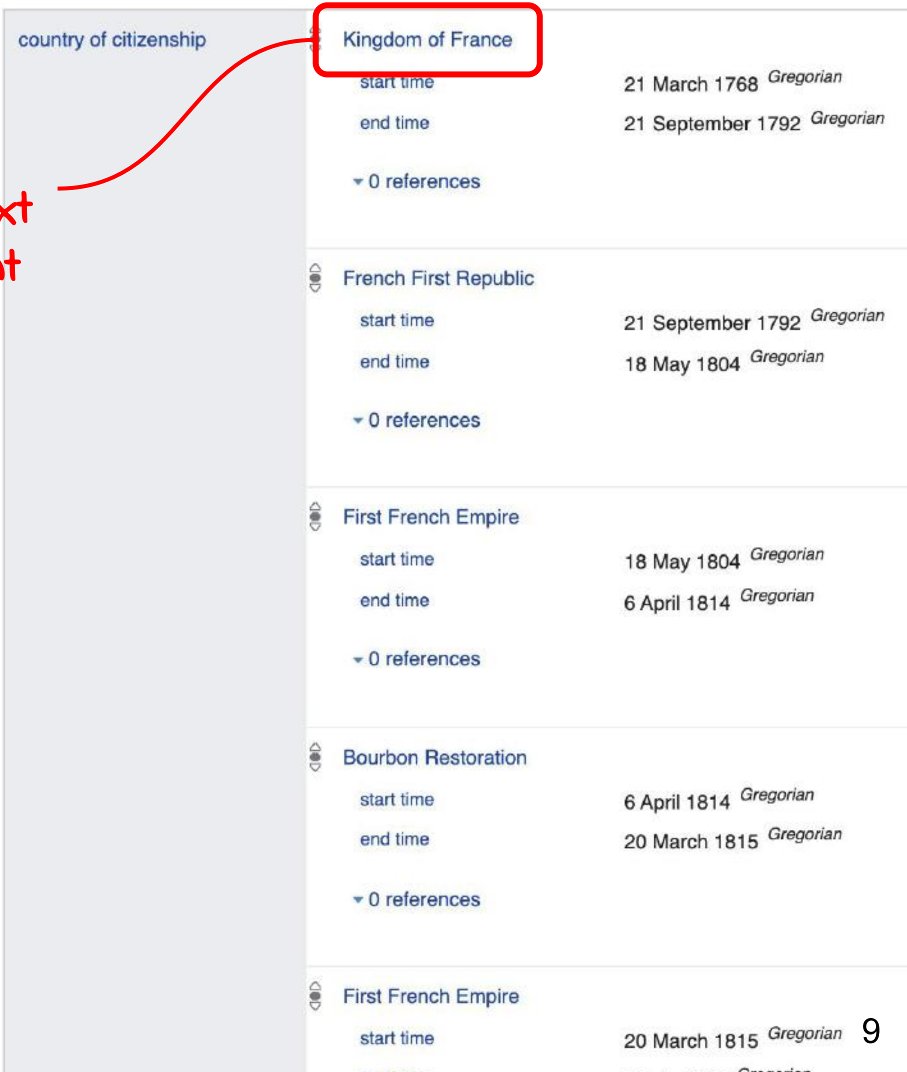


# WikiData Recap

object
=
subject/context
for statement

**different namespaces for properties**

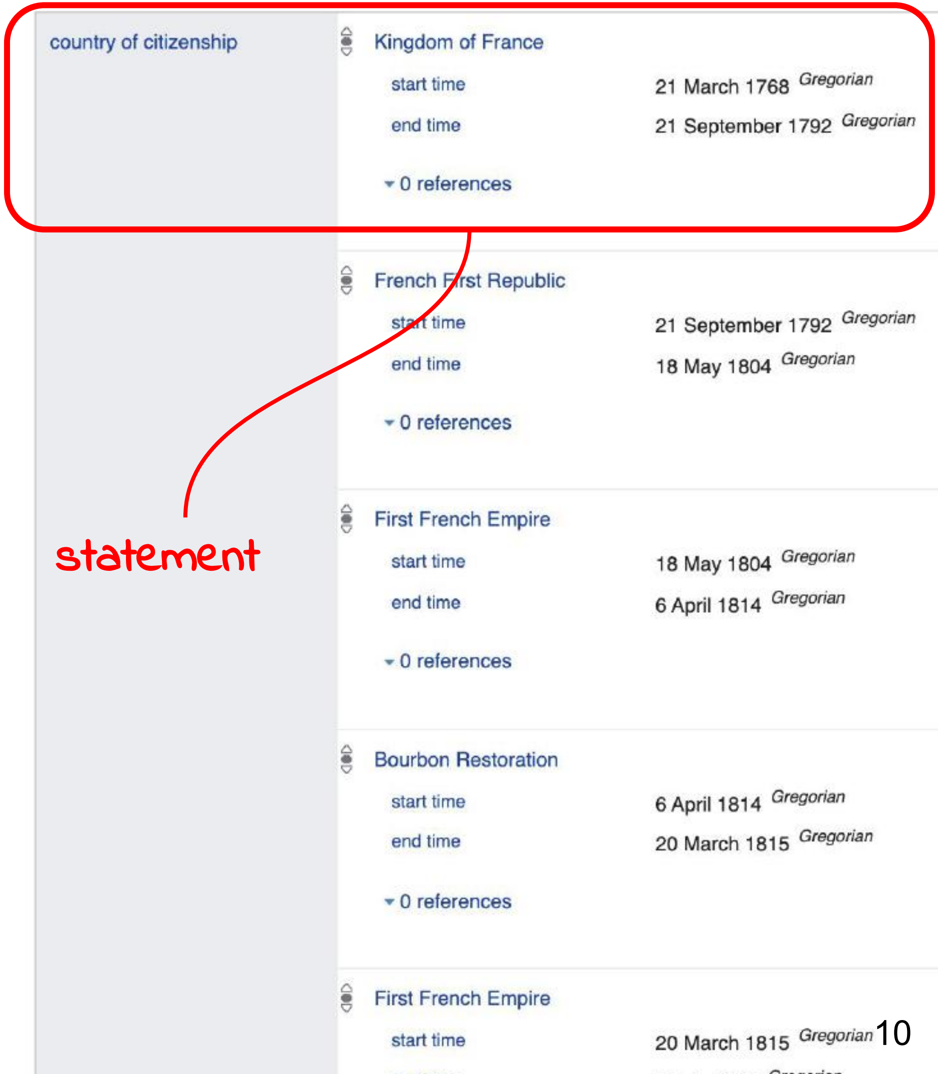
- **wdt:**
  **wd:Q8772 wdt:P27 ?country .**



# WikiData Recap

**different namespaces for properties**

- **wdt:**
  **wd:Q8772 wdt:P27 ?country .**
- **p:**
  **wd:Q8772 p:P27 ?country_statement .**

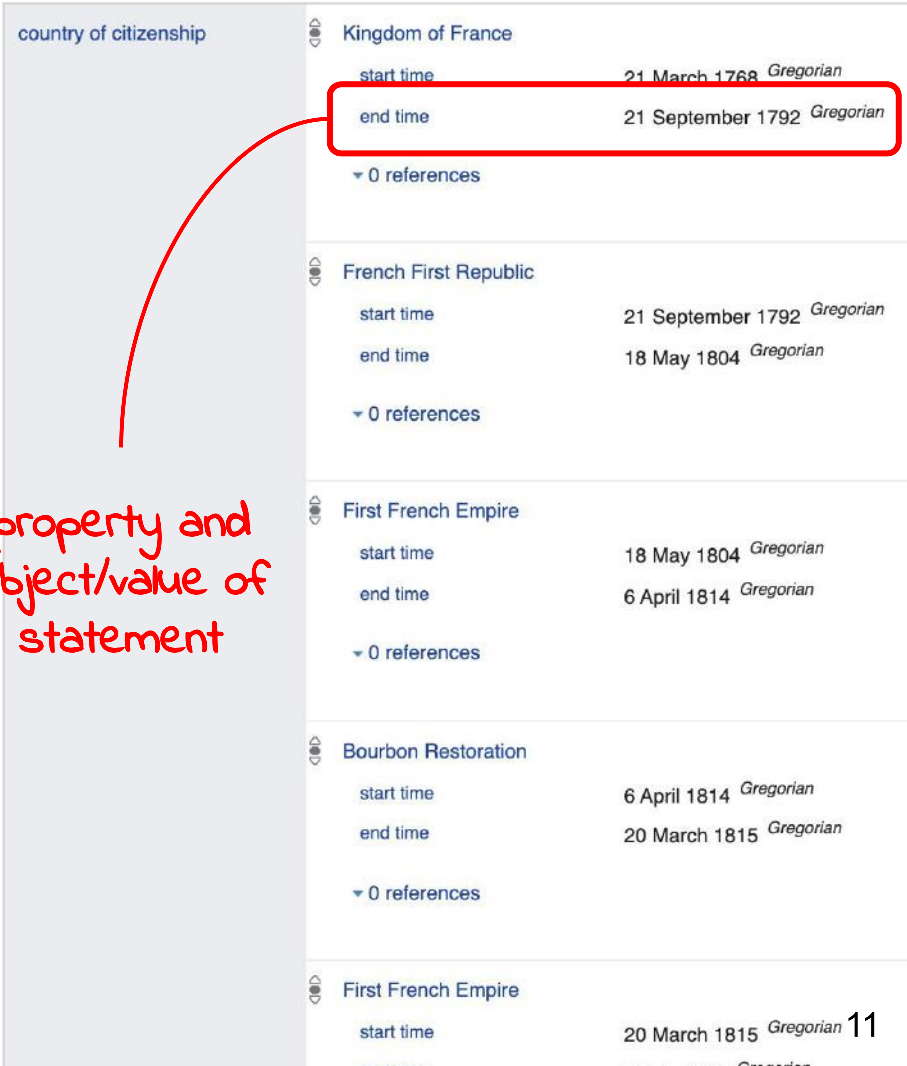


# WikiData Recap

**different namespaces for properties**

- **wdt:**
  **wd:Q8772 wdt:P27 ?country .**
- **p:**
  **wd:Q8772 p:P27 ?country_statement .**
- **pq:**
  **?country_statement pq:P582 ?statement_value**

property and object/value of statement

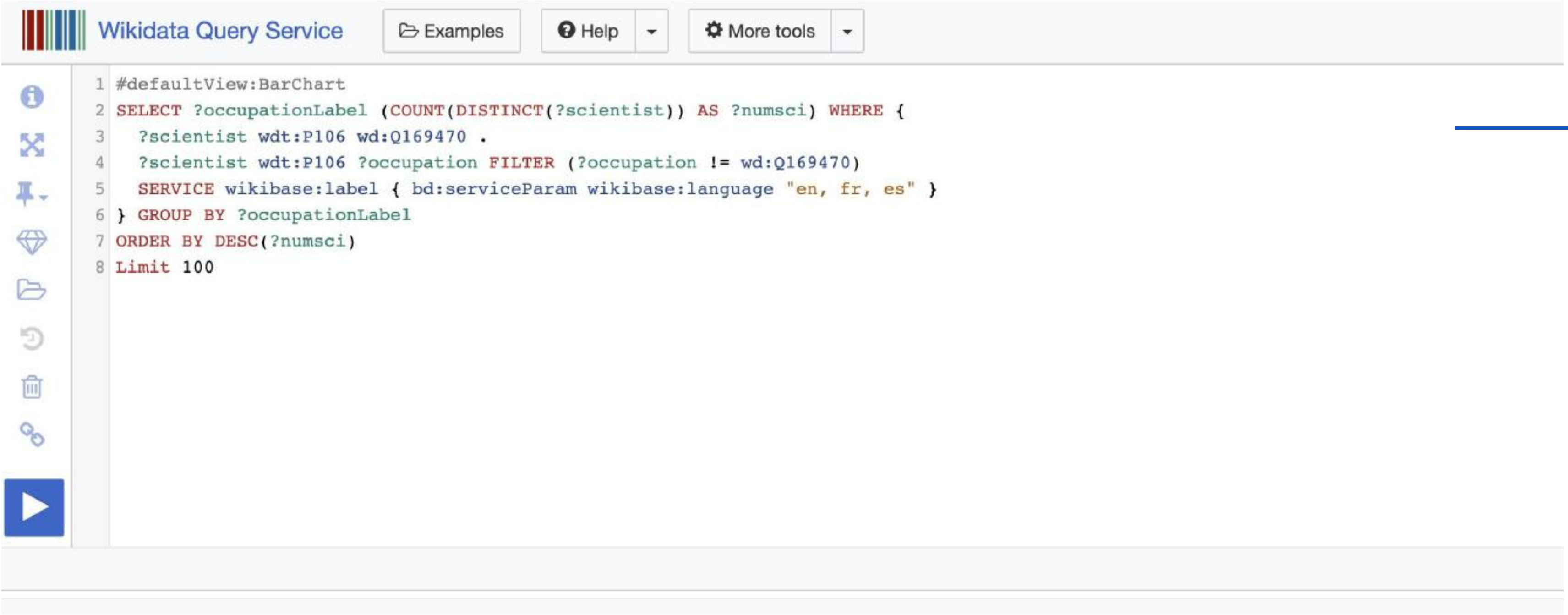


# Knowledge Graph Analytics with SPARQL

- 
- **WIKIDATA SPARQL endpoint**

# Knowledge Graph Analytics with SPARQL

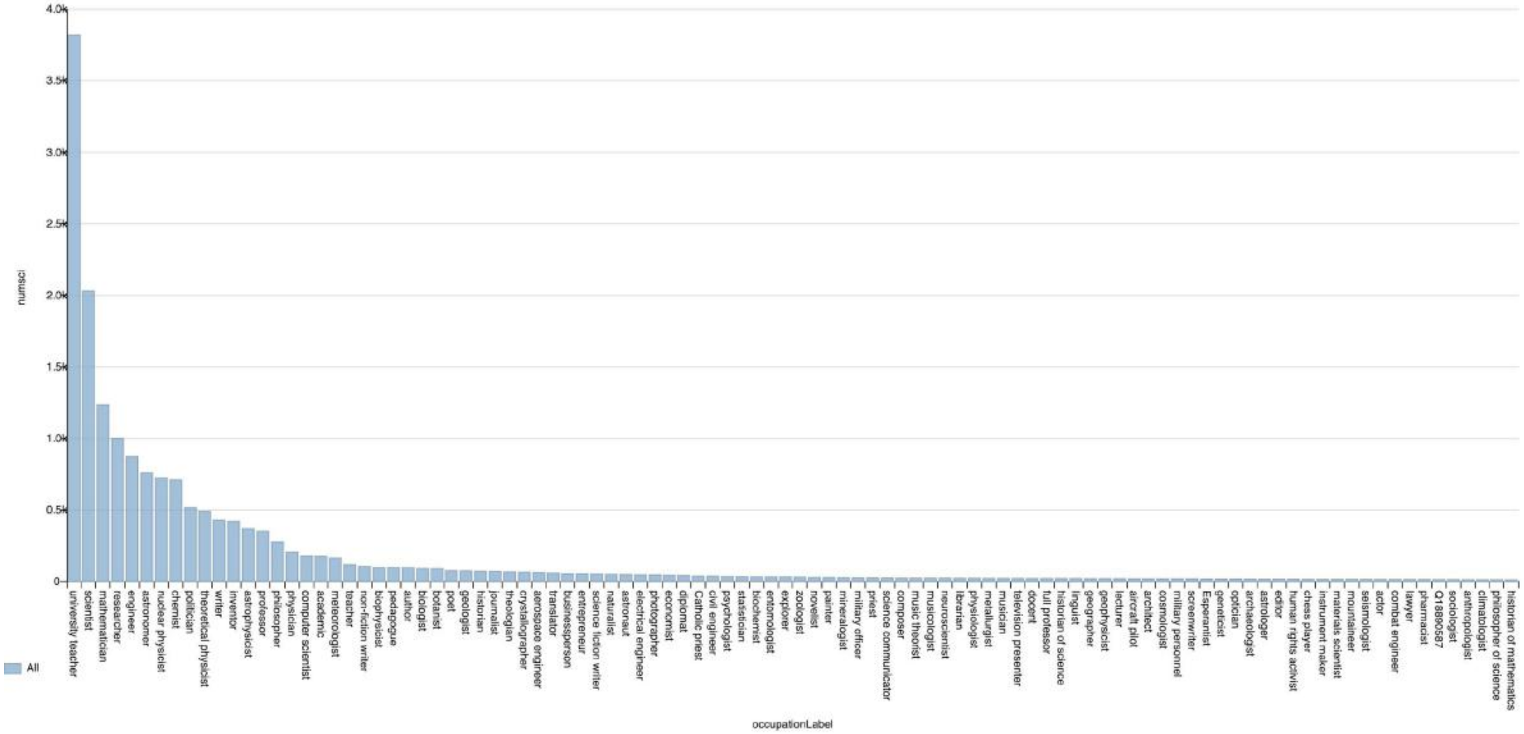
- what other occupations do Physicists have?




```
#defaultView:BarChart
SELECT ?occupationLabel (COUNT(DISTINCT(?scientist)) AS ?numsci) WHERE {
  ?scientist wdt:P106 wd:Q169470 .
  ?scientist wdt:P106 ?occupation FILTER (?occupation != wd:Q169470)
  SERVICE wikibase:label { bd:serviceParam wikibase:language "en, fr, es" }
} GROUP BY ?occupationLabel
ORDER BY DESC(?numsci)
Limit 100
```

# Knowledge Graph Analytics with SPARQL

- What other occupations do Physicists have?



# 2. Get Data

- There is so much more to know about physicists

```
SELECT ?physicist (SAMPLE(YEAR(?birthdate)) AS ?bdate)
  (COUNT(DISTINCT(?country)) AS ?countries)
  (COUNT(DISTINCT(?occupation)) AS ?occupations)
  (COUNT(DISTINCT(?employer)) AS ?employers)
  (COUNT(DISTINCT(?award)) AS ?awards)
  (COUNT(DISTINCT(?member)) AS ?members)
  (COUNT(DISTINCT(?field)) AS ?fields)
  (SAMPLE(?sex) AS ?gender)
WHERE {
  ?physicist wdt:P106 wd:Q169470 .
  ?physicist wdt:P27 ?country .
  ?physicist wdt:P569 ?birthdate .
  ?physicist wdt:P106 ?occupation .
  ?physicist wdt:P108 ?employer .
  ?physicist wdt:P166 ?award .
  ?physicist wdt:P463 ?member .
  ?physicist wdt:P21 ?sex .
  ?physicist wdt:P101 ?field .
} GROUP BY ?physicist
```

# 2. Get Data

- There is so much more to know about physicists

Wikidata Query Service | Examples | Help | More tools

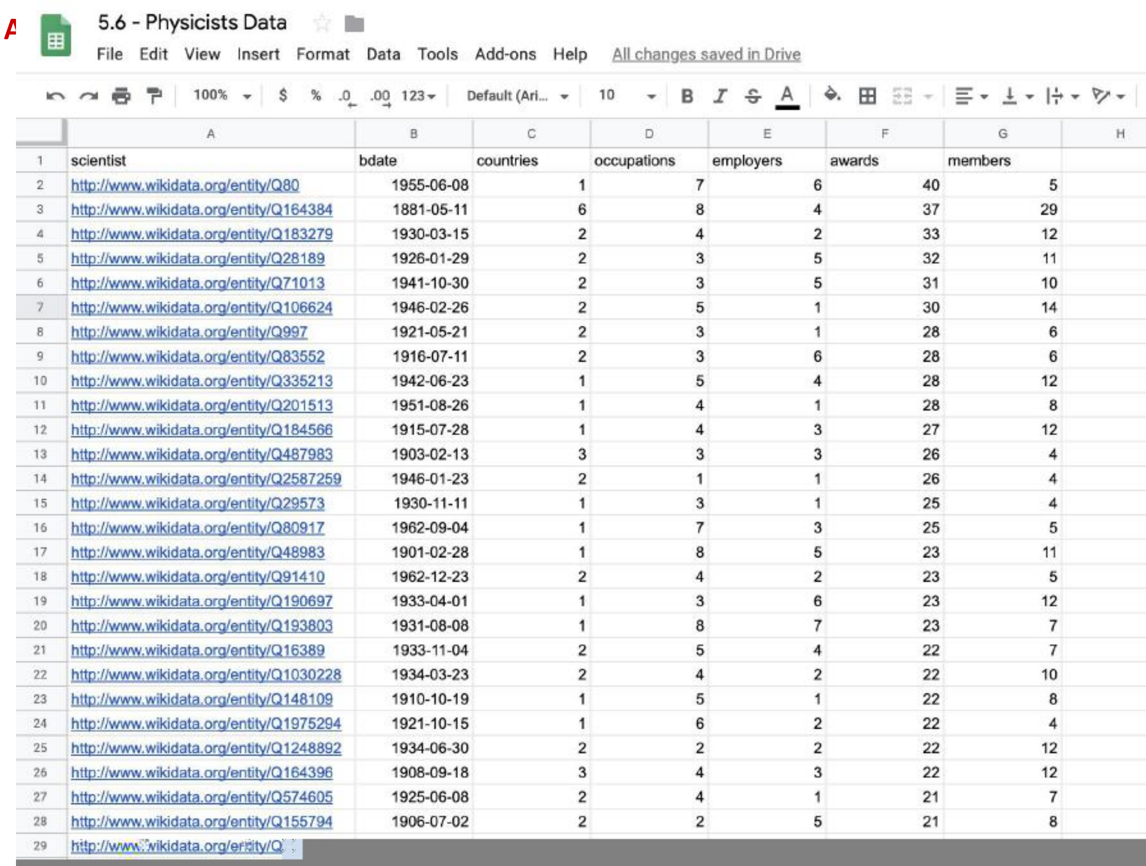
```
SELECT ?physicist (SAMPLE(YEAR(?birthdate)) AS ?bdate)
  (COUNT(DISTINCT(?country)) AS ?countries)
  (COUNT(DISTINCT(?occupation)) AS ?occupations)
  (COUNT(DISTINCT(?employer)) AS ?employers)
  (COUNT(DISTINCT(?award)) AS ?awards)
  (COUNT(DISTINCT(?member)) AS ?members)
  (COUNT(DISTINCT(?field)) AS ?fields)
  (SAMPLE(?sex) AS ?gender)
WHERE {
  ?physicist wdt:P106 wd:Q169470 .
  ?physicist wdt:P27 ?country .
  ?physicist wdt:P569 ?birthdate .
  ?physicist wdt:P106 ?occupation .
  ?physicist wdt:P108 ?employer .
  ?physicist wdt:P166 ?award .
  ?physicist wdt:P463 ?member .
  ?physicist wdt:P21 ?sex .
  ?physicist wdt:P101 ?field .
} GROUP BY ?physicist
```

1460 results in 203 ms | Code | Download

Search

| physicist | bdate | countries | occupations | employers | awards | members | fields | gender |
|---|---|---|---|---|---|---|---|---|
| wd:Q937 | 1879 | 5 | 14 | 15 | 17 | 18 | 1 | wd:Q6581097 |
| wd:Q80 | 1955 | 1 | 7 | 6 | 40 | 5 | 2 | wd:Q6581097 |
| wd:Q4517 | 1911 | 3 | 8 | 1 | 10 | 3 | 2 | wd:Q6581097 |
| wd:Q6722 | 1777 | 3 | 8 | 1 | 6 | 12 | 10 | wd:Q6581097 |
| wd:Q675 | 1775 | 1 | 4 | 2 | 2 | 10 | 1 | wd:Q6581097 |
| wd:Q680 | 1745 | 4 | 4 | 1 | 3 | 6 | 1 | wd:Q6581097 |
| wd:Q464 | 1821 | 2 | 4 | 1 | 1 | 3 | 2 | wd:Q6581097 |
| wd:Q1585 | 1789 | 2 | 3 | 2 | 3 | 4 | 1 | wd:Q6581097 |
| wd:Q30693 | 1822 | 2 | 4 | 4 | 6 | 15 | 1 | wd:Q6581097 |

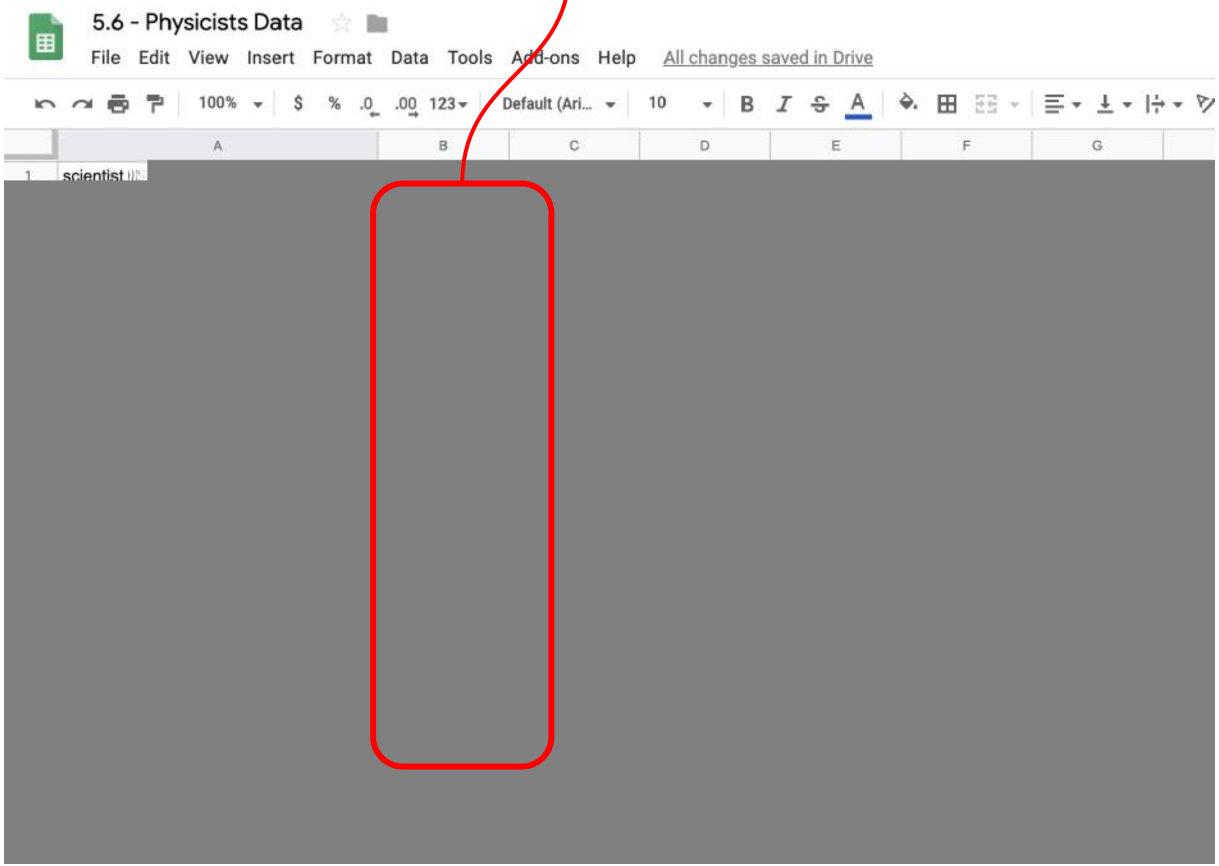
# 2. Get Data




| | scientist | bdate | countries | occupations | employers | awards | members |
|---|---|---|---|---|---|---|---|
| 2 | http://www.wikidata.org/entity/Q80 | 1955-06-08 | 1 | 7 | 6 | 40 | 5 |
| 3 | http://www.wikidata.org/entity/Q164384 | 1881-05-11 | 6 | 8 | 4 | 37 | 29 |
| 4 | http://www.wikidata.org/entity/Q183279 | 1930-03-15 | 2 | 4 | 2 | 33 | 12 |
| 5 | http://www.wikidata.org/entity/Q28189 | 1926-01-29 | 2 | 3 | 5 | 32 | 11 |
| 6 | http://www.wikidata.org/entity/Q71013 | 1941-10-30 | 2 | 3 | 5 | 31 | 10 |
| 7 | http://www.wikidata.org/entity/Q106624 | 1946-02-26 | 2 | 5 | 1 | 30 | 14 |
| 8 | http://www.wikidata.org/entity/Q997 | 1921-05-21 | 2 | 3 | 1 | 28 | 6 |
| 9 | http://www.wikidata.org/entity/Q83552 | 1916-07-11 | 2 | 3 | 6 | 28 | 6 |
| 10 | http://www.wikidata.org/entity/Q335213 | 1942-06-23 | 1 | 5 | 4 | 28 | 12 |
| 11 | http://www.wikidata.org/entity/Q201513 | 1951-08-26 | 1 | 4 | 1 | 28 | 8 |
| 12 | http://www.wikidata.org/entity/Q184566 | 1915-07-28 | 1 | 4 | 3 | 27 | 12 |
| 13 | http://www.wikidata.org/entity/Q487983 | 1903-02-13 | 3 | 3 | 3 | 26 | 4 |
| 14 | http://www.wikidata.org/entity/Q2587259 | 1946-01-23 | 2 | 1 | 1 | 26 | 4 |
| 15 | http://www.wikidata.org/entity/Q29573 | 1930-11-11 | 1 | 3 | 1 | 25 | 4 |
| 16 | http://www.wikidata.org/entity/Q80917 | 1962-09-04 | 1 | 7 | 3 | 25 | 5 |
| 17 | http://www.wikidata.org/entity/Q48983 | 1901-02-28 | 1 | 8 | 5 | 23 | 11 |
| 18 | http://www.wikidata.org/entity/Q91410 | 1962-12-23 | 2 | 4 | 2 | 23 | 5 |
| 19 | http://www.wikidata.org/entity/Q190697 | 1933-04-01 | 1 | 3 | 6 | 23 | 12 |
| 20 | http://www.wikidata.org/entity/Q193803 | 1931-08-08 | 1 | 8 | 7 | 23 | 7 |
| 21 | http://www.wikidata.org/entity/Q16389 | 1933-11-04 | 2 | 5 | 4 | 22 | 7 |
| 22 | http://www.wikidata.org/entity/Q1030228 | 1934-03-23 | 2 | 4 | 2 | 22 | 10 |
| 23 | http://www.wikidata.org/entity/Q148109 | 1910-10-19 | 1 | 5 | 1 | 22 | 8 |
| 24 | http://www.wikidata.org/entity/Q1975294 | 1921-10-15 | 1 | 6 | 2 | 22 | 4 |
| 25 | http://www.wikidata.org/entity/Q1248892 | 1934-06-30 | 2 | 2 | 2 | 22 | 12 |
| 26 | http://www.wikidata.org/entity/Q164396 | 1908-09-18 | 3 | 4 | 3 | 22 | 12 |
| 27 | http://www.wikidata.org/entity/Q574605 | 1925-06-08 | 2 | 4 | 1 | 21 | 7 |
| 28 | http://www.wikidata.org/entity/Q155794 | 1906-07-02 | 2 | 2 | 5 | 21 | 8 |
| 29 | http://www.wikidata.org/entity/Q[illegible] | | | | | | |

# 3. Clean Up Data

Can we trust this data?

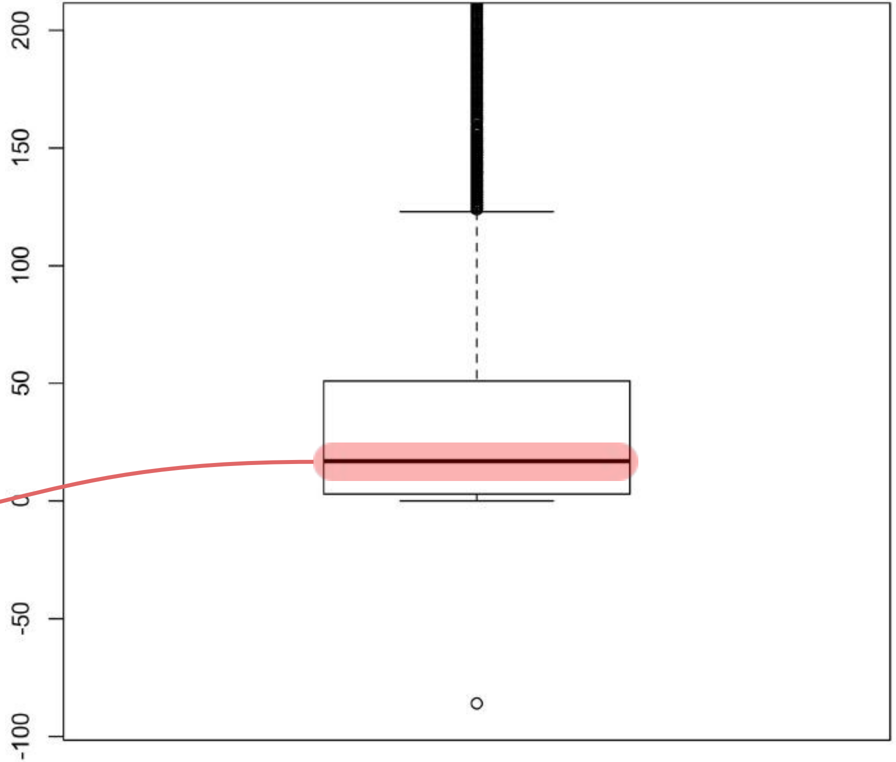


# Excursion: Learning How to Read Box Plots



**median**

```
      X0
Min.   : -86.00
1st Qu.:   3.00
Median :  17.00
Mean   :  38.85
3rd Qu.:  51.00
Max.   :1329.00
```

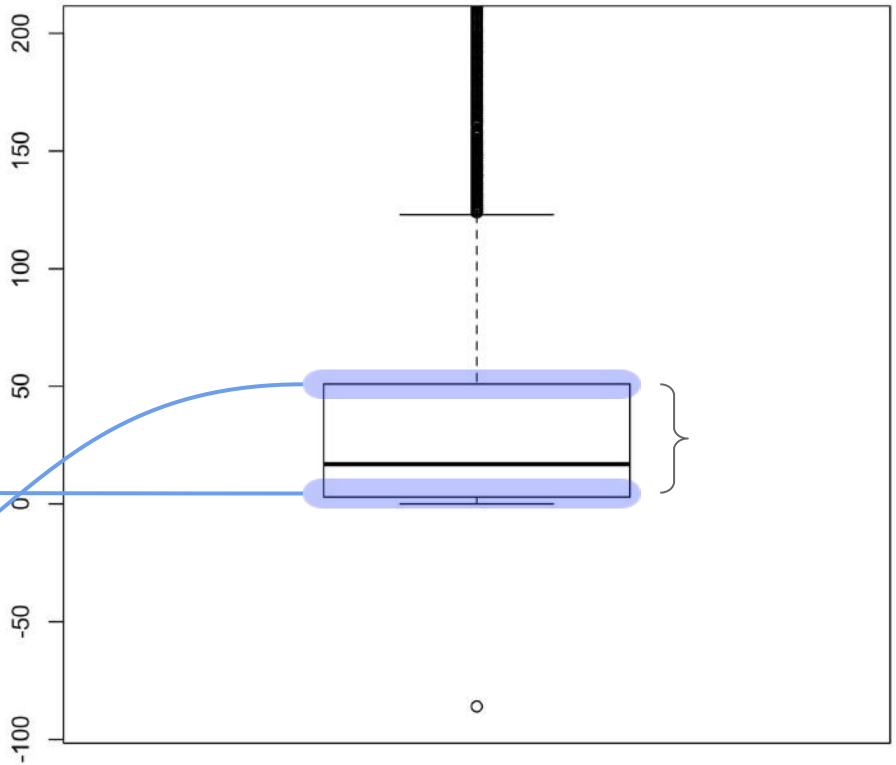


# Excursion: Learning How to Read Box Plots

**first quartile ($Q_1$)**

**third quartile ($Q_3$)**

```
      X0
Min.   : -86.00
1st Qu.:   3.00
Median :  17.00
Mean   :  38.85
3rd Qu.:  51.00
Max.   :1329.00
```



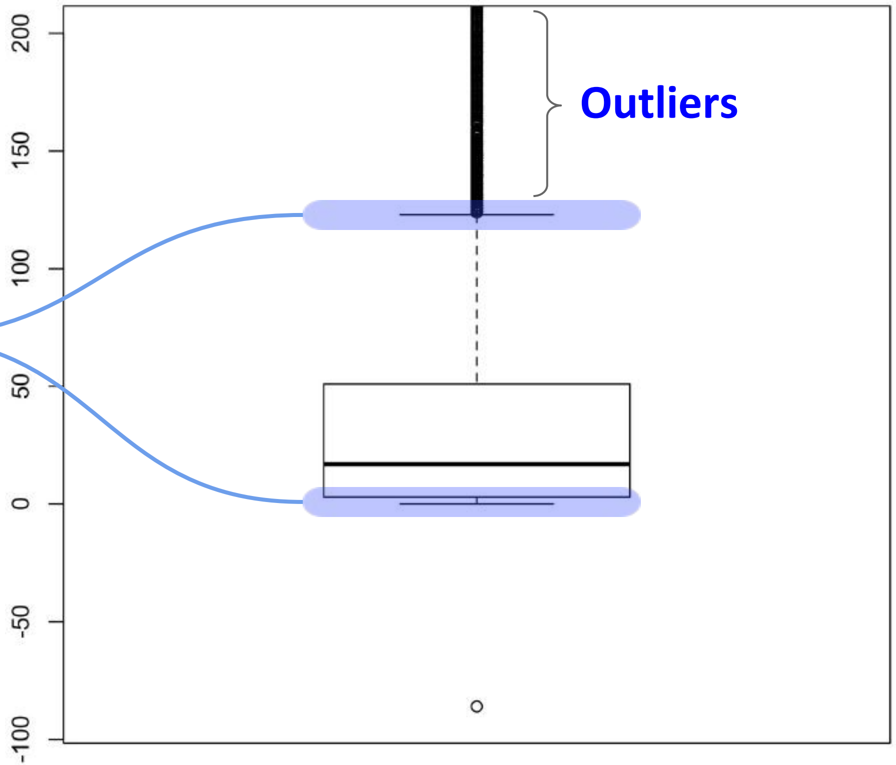
# Excursion: Learning How to Read Box Plots

**Whiskers**

**outlier**

**Whiskers**

```
        X0
Min.   : -86.00
1st Qu.:   3.00
Median :  17.00
Mean   :  38.85
3rd Qu.:  51.00
Max.   :1329.00
```

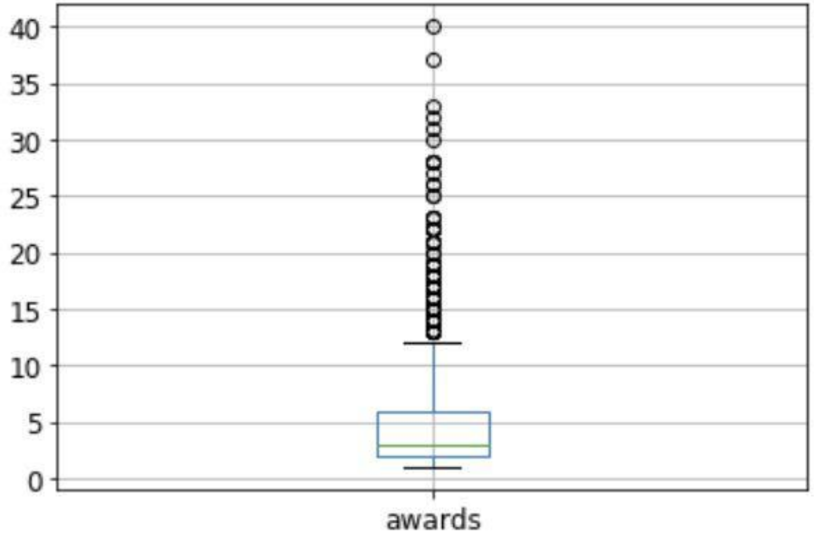


# Excursion: Learning How to Read Box Plots

## Analyse the Data

```
[29] physicists.boxplot('awards')
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f3c00578f28>
```



# Advanced Knowledge Graph Applications

## Next Lecture...

## Picture References:

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